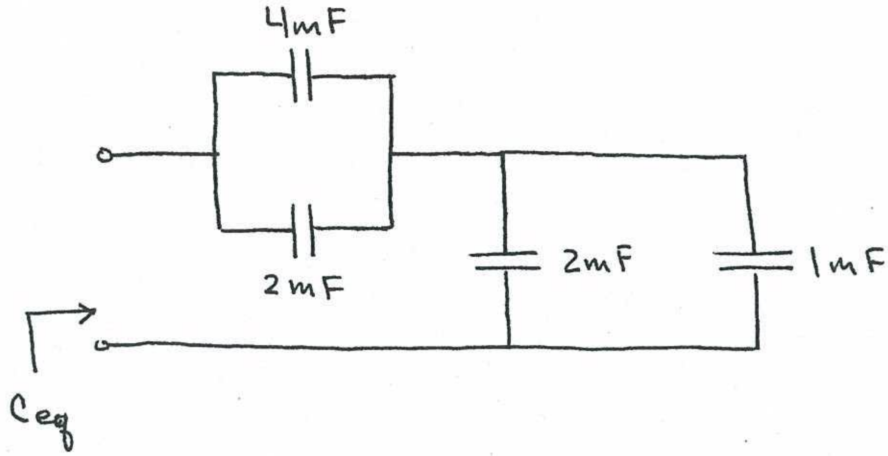
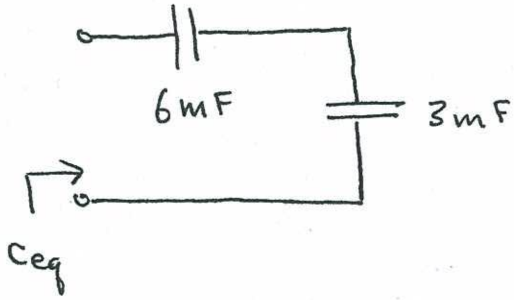
6.1



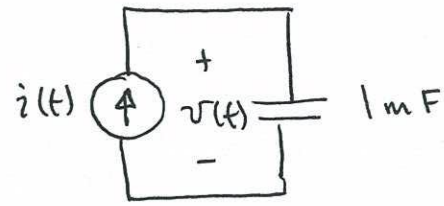
Capacitors in parallel add directly, so the above is equivalent to:

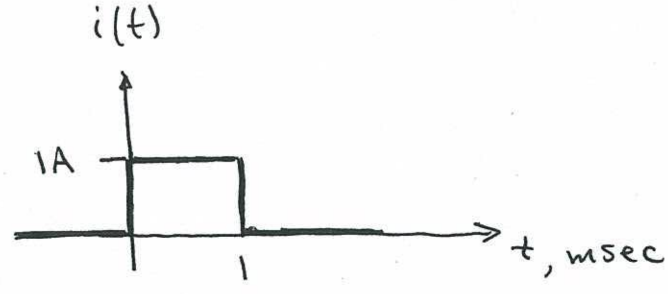


Capacitors in series add like resistors in parallel:

$$C_{eq} = \frac{(3mF)(6mF)}{3mF + 6mF} \Rightarrow \boxed{C_{eq} = 2mF}$$

6.2



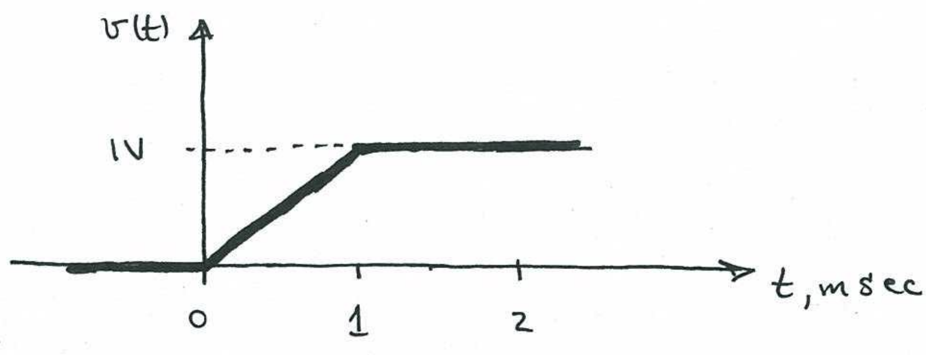


voltage - current relation for capacitors:

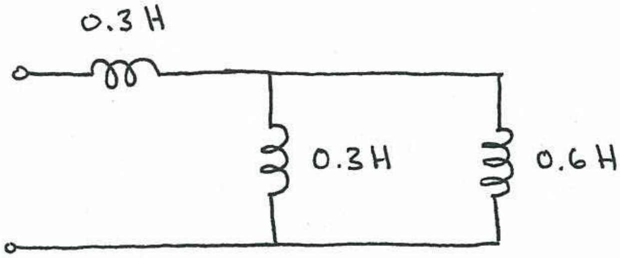
$$v(t) = \frac{1}{C}\int_0^t i(t)\,dt + \cancel{v(0)}$$

0 (capacitor initially uncharged)

$$v(t) = 1000 \int_0^t 1\,dt\,, \quad 0 < t < 1 \times 10^{-3}\ \text{sec}$$



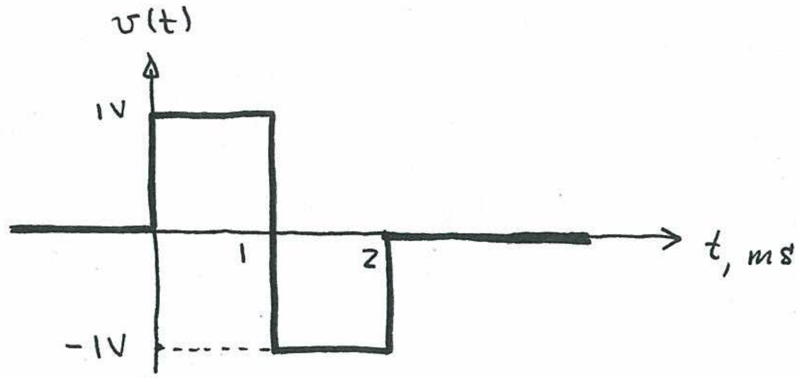
6.3



$$L_{eq} = 0.3\,H + \frac{(0.3\,H)(0.6\,H)}{0.3\,H + 0.6\,H}$$

$$\boxed{L_{eq} = 0.5\,H}$$

6.4



$L = 1\,mH$

Voltage-current relationship:

$$i(t) = \frac{1}{L}\int_0^t v(t)\,dt + \cancel{i(t=0)}^{\,0}$$

0 ← assume initial current is zero (inductor relaxed at time t=0).